

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Duration :60 Mins

Total Marks:50

Assessment Mode: In Class

Annexure A

CONCEPT MAPPING

Code of Conduct:

I _K. Madhulika(192524190)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Madhulika

Q. No	Understanding of Data (25)	Analysis (25)	Application of Concepts (20)	Conclusion (20)	Clarity (10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 25	/ 20	/ 20	/ 10	/ 100

Annexure A

CONCEPT MAPPING

1. Procedure Calls in Intermediate Code
2. Parameter Passing in Intermediate Code
3. Activation Records and Procedure Calls
4. Intermediate Code for Function Calls
5. Intermediate Code for Return Statements
6. Intermediate Code Generation for Nested Procedures
7. Syntax-Directed Translation for Intermediate Code
8. Intermediate Code Generation Using Syntax-Directed Definitions
9. Complete Intermediate Code Generation Workflow
10. Intermediate Code Generation – Overview and Need
11. Intermediate Languages and Their Types
12. Intermediate Representation in Compiler Design
13. Three-Address Code and Its Structure
14. Quadruples, Triples and Indirect Triples
15. Intermediate Code for Declarations
16. Translation of Variable Declarations
17. Type Information in Intermediate Code
18. Intermediate Code for Assignment Statements
19. Translation of Arithmetic Assignment Expressions
20. Intermediate Code for Array Assignments
21. Intermediate Code for Boolean Expressions
22. Short-Circuit Evaluation of Boolean Expressions
23. Translation of Relational Expressions
24. Translation of Logical Operators
25. Boolean Expressions Using Three-Address Code
26. Control Flow in Intermediate Code
27. Intermediate Code for Conditional Statements
28. Intermediate Code for If–Else Statements
29. Intermediate Code for While Statements
30. Intermediate Code for For Statements
31. Intermediate Code for Case Statements

32. Translation of Switch–Case Constructs
33. Control Flow Graph for Case Statements
34. Backpatching – Concept and Need
35. Backpatching Lists and Their Operations
36. True and False Lists in Backpatching
37. Backpatching for Boolean Expressions
38. Backpatching for Conditional Statements
39. Backpatching for Loops
40. Backpatching and Control Flow

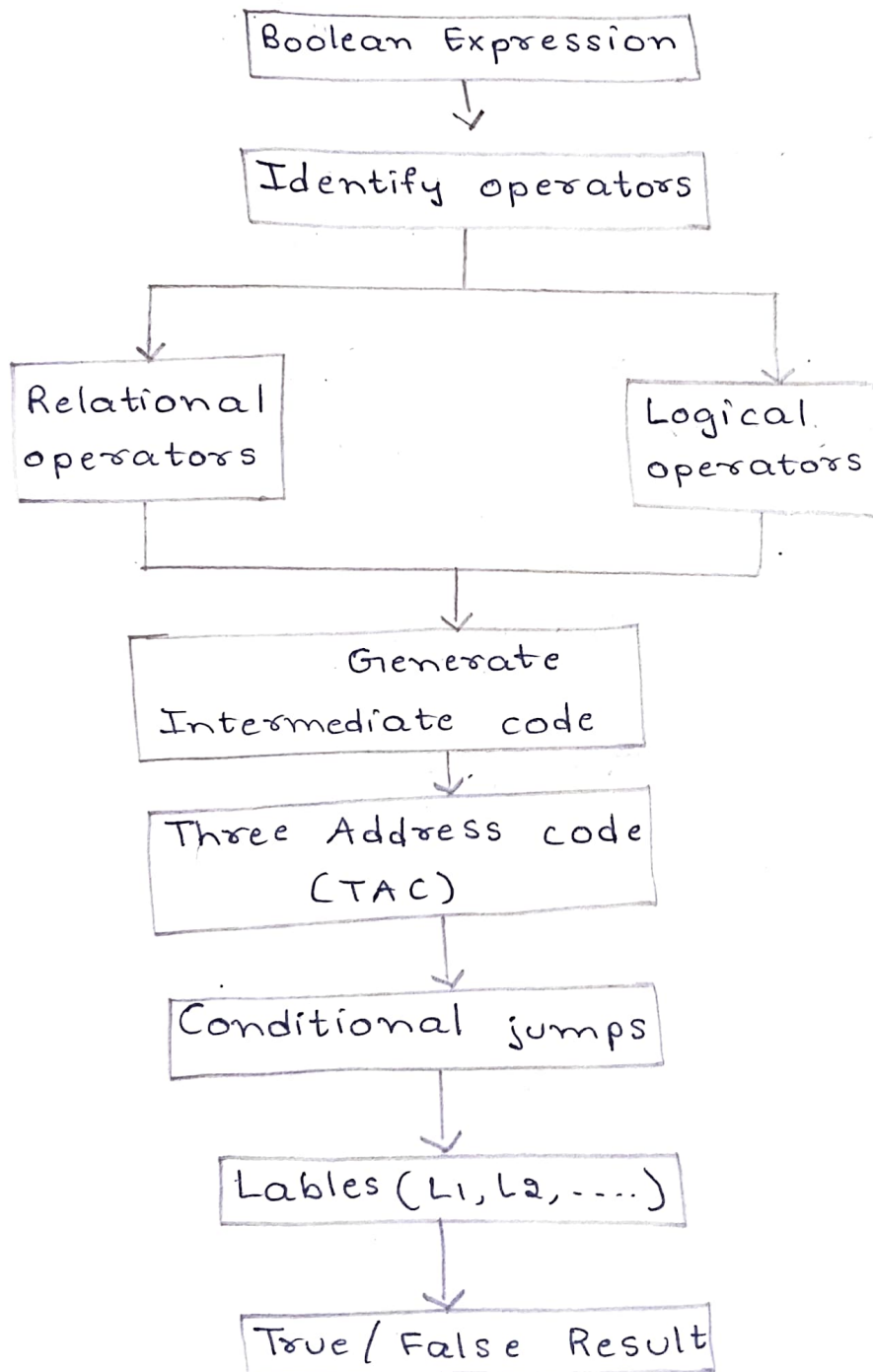
CO4 - AT3

CO4 AT3 - Concept Mapping

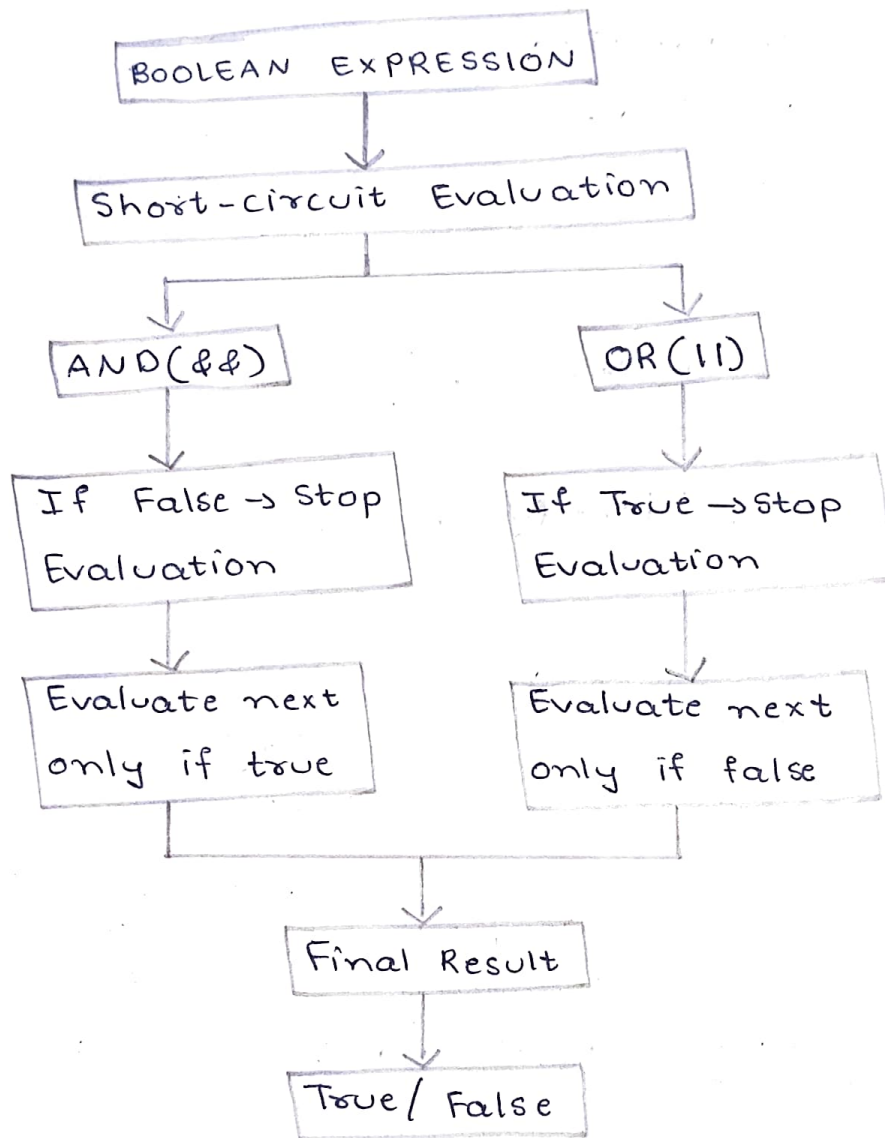
CSA 1419 - Compiler Design

Name : K. Madhulika

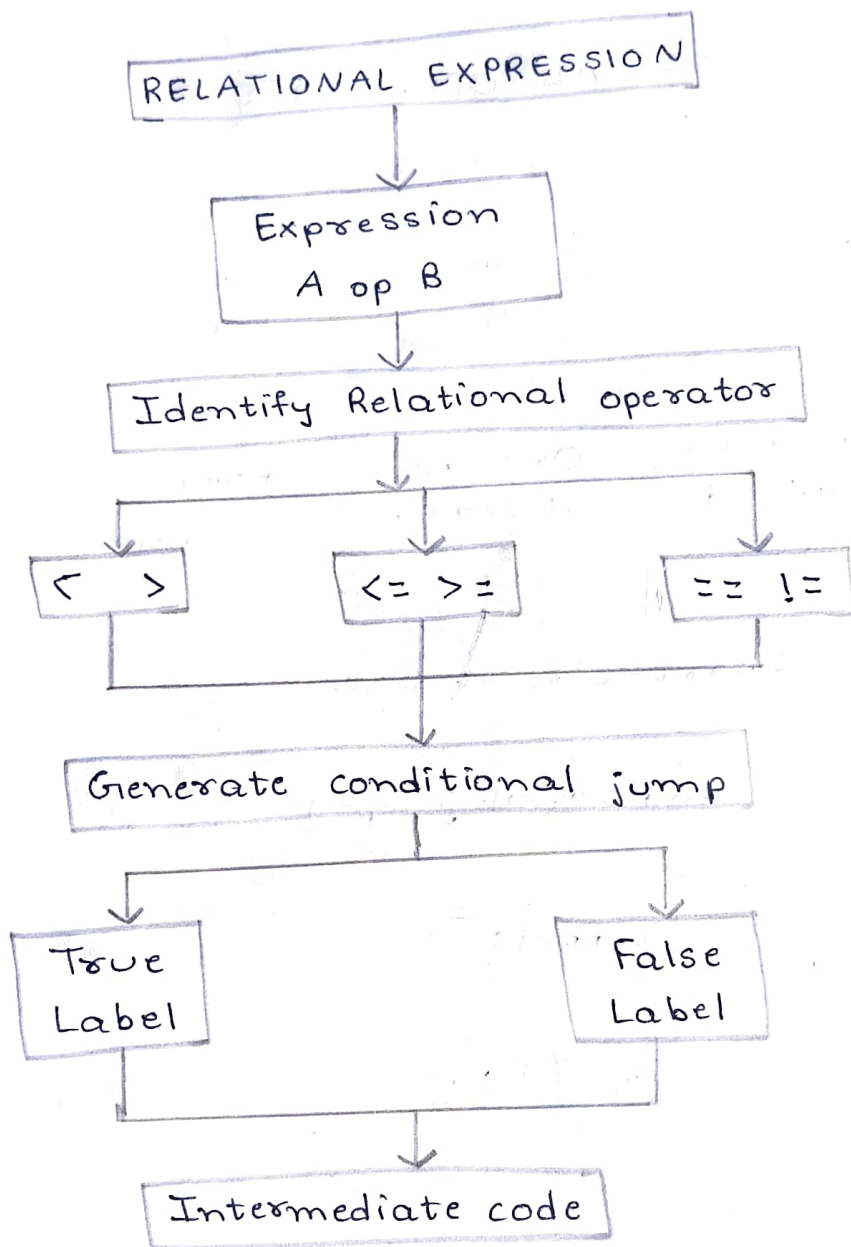
Reg no : 192524190

21. Intermediate code for boolean expressions:

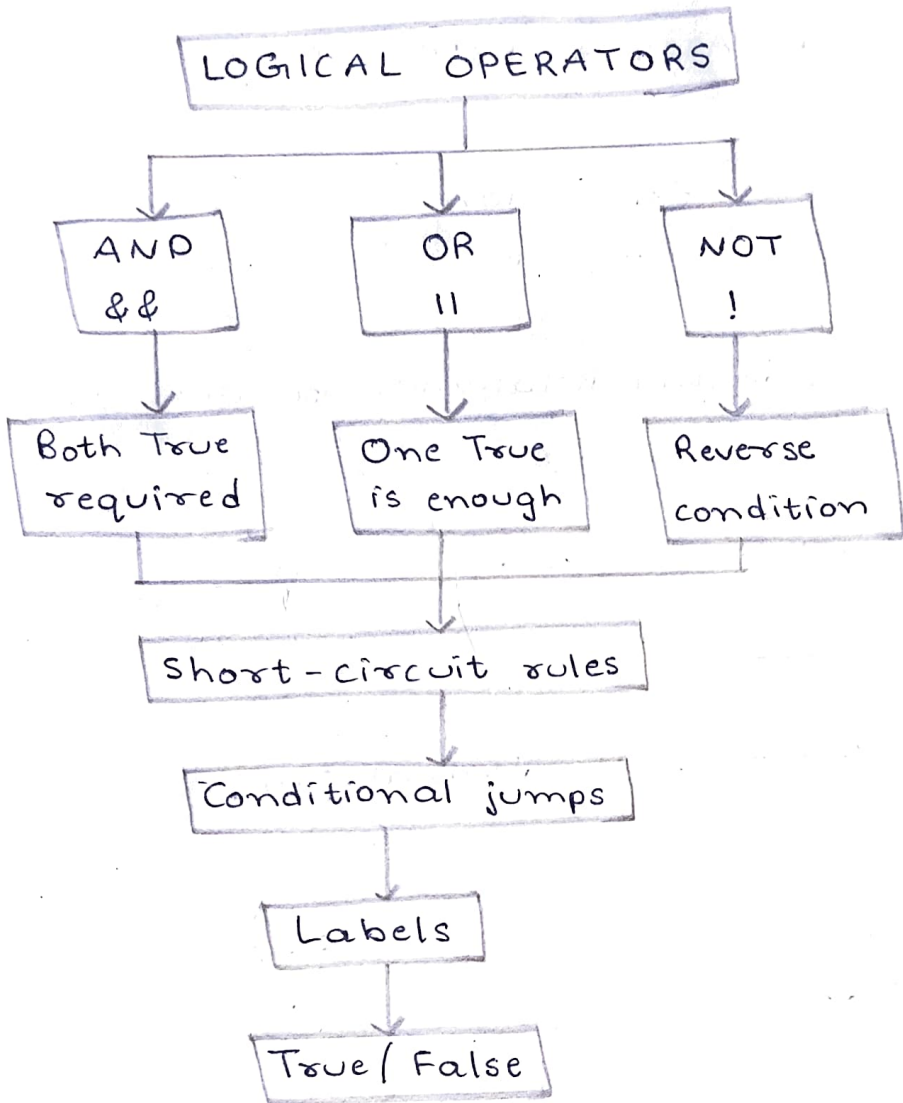
22. Short circuit Evaluation of Boolean expressions:



23. Translation of Relational Expressions:



24. Translation of Logical operators:



25. Boolean Expressions using Three - Address code :

